MOBILE COMPUTING

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

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TASK 1 – Design Alternative Frequency-Planning Strategies (5 Marks)

A fixed spectrum can serve many cells by reusing the same frequencies at sufficiently separated locations. The main constraint is co-channel interference: cells using the same channel set must be separated far enough to maintain the required SIR. Three distinct strategies are evaluated.

Strategy A – Fixed cluster frequency reuse

The township is divided into hexagonal cells. The 336-channel block is divided into N channel groups and the groups are assigned repeatedly in a regular pattern. For N = 9, the cluster can be generated by i = 3 and j = 0, because $N = i^2 + ij + j^2 = 9$. Each cell is assigned one group. The same group appears again after the reuse distance $D = R\sqrt{(3N)}$.

Access concept: in a traditional FDMA system, each duplex channel is a paired uplink/downlink frequency. In TDMA, a carrier is divided into time slots; in OFDMA, subcarriers and resource blocks are scheduled. The planning abstraction here counts 336 duplex traffic-channel equivalents. Control channels are reserved separately as stated in the question; the traffic-channel pool is used for the Erlang calculation.

Design principle: spatial reuse. Assumptions: regular hexagonal layout, approximately uniform traffic, a common reuse pattern, and adequate frequency separation. It is worth investigating because it is simple, predictable and provides a baseline for capacity and SIR.

```
INPUT: total_channels S=336, candidate cluster sizes N
FOR each valid (i,j) pair:
    N = i*i + i*j + j*j
    channels_per_cell = floor(S / N)
    Q = sqrt(3*N)
    estimate SIR for n=2.5, 3 and 4
    reject N if minimum SIR < 12 dB
    calculate Erlang-B capacity at 2% blocking
    calculate cells required for forecast traffic
SELECT feasible N with lowest lifecycle cost
ASSIGN N channel groups to the hexagonal grid
REPEAT the N-group pattern across the township
```

Strategy B – Sectorised frequency reuse

Each base station uses three directional antennas, each covering a 120° sector. The same cluster reuse method is retained, but the directional antenna pattern suppresses many first-tier interferers. The simplified first-tier model reduces the dominant interferer count from $I_0 = 6$ to approximately $I_0 = 2$. For N = 9, 36 of the 37 nominal channels per cell are divided into three groups of 12 channels, one group per sector.

Design principle: spatial filtering. It is worth investigating because it improves SIR without increasing the licensed spectrum and can support higher traffic density. Assumptions include correct antenna orientation, adequate front-to-back ratio, and manageable traffic balance between sectors.

```
FOR each cell:
    divide cell into 3 sectors of 120 degrees
    assign 12 traffic channels to each sector
    orient antennas toward the sector service area
    estimate SIR using about 2 dominant first-tier interferers
    if any sector exceeds 2% blocking:
        add capacity by splitting the cell or adding a microcell
    retain the reuse pattern and verify edge SIR
```

Strategy C – Dynamic channel assignment / borrowing with hierarchical cells

A macro layer provides continuous township coverage. Microcells are added at high-demand locations such as hostels, the library, academic blocks and the stadium. A cell normally uses its planned channel set; when demand rises, it may borrow an idle channel from a neighbouring cell only if the donor remains below its blocking limit and the borrowed channel does not violate the SIR/guard-distance rule. A guard-channel policy can protect handoff calls.

Design principles: trunking efficiency, adaptive resource allocation and hierarchical spatial reuse. This strategy is worth investigating because demand is not uniform and changes by time of day. It can use idle channels more efficiently than a rigid assignment, but coordination and interference checks are essential.

```
FOR each call request in cell c:
    first try an idle channel in c's home set
    IF none is available:
```

search neighbouring cells for an idle borrowable channel

check donor-cell reserve and co-channel SIR constraint

IF a safe channel exists:

borrow channel and record its owner

ELSE:

block the call

WHEN call ends or channel is no longer needed:

return borrowed channel to its home cell

AT busy hotspots:

recursively add low-power microcells and re-plan local channels

Preprocessing and who bears the cost

- Traffic map: estimate subscribers and busy-hour Erlangs by academic blocks, hostels, residential colonies, roads and stadium. The operator's planning team collects or purchases the demand data.
- Propagation survey: measure received signal strength, path loss, shadowing, building penetration and handoff points. The operator bears the drive-test and measurement cost.
- Hex-grid and site survey: map candidate rooftops/poles, power, backhaul, access, zoning and safety constraints. The operator and site owner share the site-acquisition effort.
- Channel and antenna planning: prepare frequency groups, sector azimuths, tilt, power and neighbour lists. This is an operator radio-planning and optimisation cost.

Conclusion: fixed reuse is the baseline; sectorisation is the strongest spectrum-preserving improvement; dynamic borrowing and macro–micro layering are most useful for non-uniform and temporary demand.

TASK 2 – Identify the Governing Design Parameter (2 Marks)

Strategy	Governing quantity	Why it governs	Growth response
Fixed reuse	Cluster size N and reuse ratio Q = D/R = $\sqrt{(3N)}$	N determines channels per cell (≈ S/N) and co-channel separation. Increasing N improves SIR but reduces channels per cell.	Keep the smallest N that meets SIR; add cells / reduce R as traffic grows.
Sectorised reuse	Interferer count I_0 and sector channel count	Sector antenna pattern reduces I_0; channels per sector determine Erlang-B capacity and blocking.	Maintain sector SIR; split or add when a sector becomes overloaded.

Dynamic borrowing / hierarchy	Borrowable-channel pool and cell radius R	Borrowing improves trunking but is limited by interference and donor reserve; R controls coverage and spatial reuse.	Increase microcell density at hotspots; borrow only after SIR and guard checks.

As subscribers rise from 1,000 to 100,000, offered traffic increases linearly from 25 to 2,500 Erlangs. N cannot be reduced below the interference-safe value merely to obtain more channels. Instead, the network must increase the number of cells, reduce cell radius, use sectorisation, and deploy microcells in concentrated demand areas. The most important system-level quantity becomes the number of cells M that can reuse the same fixed spectrum while preserving SIR.

TASK 3 – Derive the Capacity and Interference Functions Mathematically (6 Marks)

1. Hexagonal geometry and cluster size

For a regular hexagon with cell radius R (centre to vertex), the cell area is six equilateral triangles of side R:

$$A_{cell} = 6 \times (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598R^2 \text{ km}^2.$$

If the service area is $A_T = 12$ km², the approximate number of cells needed for coverage is:

$$M_{cover} = \text{ceil}[12 / (2.598R^2)].$$

A valid hexagonal cluster size is obtained by moving i cells along one chain of hexagons, turning 60°, and moving j cells. The number of cells in a cluster is:

$$N = i^2 + ij + j^2, \qquad Q = D/R = \sqrt{(3N)}, \qquad D = R\sqrt{(3N)}.$$

For the selected N = 9, i = 3 and j = 0. Therefore $Q = \sqrt{27} = 5.196$ and $D = 5.196R$.

2. Channels per cell and Erlang-B capacity

The average offered traffic is subscriber count multiplied by traffic per subscriber:

$$A_{total} = U \times 0.025 \text{ Erlangs}.$$

For N = 9, floor(336/9) = 37 nominal channels per cell. To provide three equal sector pools, 36 are assigned (12 per sector) and one is kept as a spare/guard resource. The Erlang-B recursion is:

$$B(0,A) = 1; \qquad B(m,A) = [A\, B(m-1,A)] / [m + A\, B(m-1,A)].$$

Solving B(12,A) = 0.02 gives A_sector ≈ 6.615 Erlangs. Three sectors therefore provide:

C_cell = 3 × 6.615 = 19.845 Erlangs/cell (approximately).

This is a conservative sector-by-sector calculation: it does not assume that unused capacity in one sector can always be transferred to another. For M cells, total traffic capacity is approximately:

C_system = M × C_cell, M_required = ceil(A_total / C_cell).

3. Required cells and radius for subscriber growth

Subscribers	Offered traffic	Cells for capacity	Selected cells	Approx. R from 12 km²
1,000	25 Erlangs	2	10 (coverage floor)	0.679 km
10,000	250 Erlangs	13	13	0.596 km
20,000	500 Erlangs	26	26	0.421 km
100,000	2,500 Erlangs	126	126	0.191 km

The 1,000-subscriber case is coverage-limited rather than capacity-limited. For larger cases the capacity requirement determines the number of cells. Radius is estimated from R = √[12/(2.598M)]. Actual site placement will not form a perfect hexagonal tessellation.

4. Co-channel SIR and sectoring

With path-loss exponent n and I_0 equal-strength first-tier interferers, the standard approximation is:

SIR ≈ (D/R)^n / I_0 = (3N)^(n/2) / I_0.

In decibels:

SIR_dB ≈ 10 $\log_{10}$[(3N)^(n/2) / I_0].

For omnidirectional cells I_0 ≈ 6. For three 120° sectors, a simplified dominant-interferer model uses I_0 ≈ 2. For a conservative edge estimate, the nearest co-channel interferer is approximately D−R away:

SIR_edge ≈ ([(D−R)/R]^n) / I_0 = (√(3N)−1)^n / I_0.

Case	N	n	I_0	SIR estimate
Omnidirectional standard approximation	9	2.5	6	10.11 dB

Sectorised standard approximation	9	2.5	2	14.88 dB
Sectorised conservative cell-edge estimate	9	2.5	2	12.56 dB
Sectorised conservative cell-edge estimate	9	4	2	21.43 dB
Sanity check requested in assignment	7	4	6	18.66 dB ≈ 18.7 dB

The selected design uses the conservative edge estimate of 12.56 dB for n = 2.5, slightly above the 12 dB threshold. Real antenna patterns, terrain and shadowing must be measured; if the measured 5th-percentile SIR is below 12 dB, the operator should adjust tilt/power, add local sites or use a larger local reuse distance.

5. Cost–capacity relationship

$$C_{total}(M,N,s) = M \times s \times A_B(m_s, 0.02),$$

where s is the number of sectors per cell and m_s is the number of traffic channels per sector. For this design, s = 3 and m_s = 12. A simple lifecycle cost model is:

$$C_{life} = C_{survey} + M(C_{site} + C_{radio} + C_{backhaul}) + C_{optimisation} + C_{maintenance}.$$

The cost per supported Erlang is C_{life}/C_{total}. The assignment gives no rupee prices, so quoting a monetary total would be fabricated. A valid procurement estimate should obtain local quotes for survey, civil work, radio equipment, power, backhaul, rent, maintenance and temporary event equipment. Increasing M raises site and maintenance cost, while sectorisation improves SIR and capacity without purchasing additional spectrum.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour (4 Marks)

SIR depends on mobile position because the desired base-station distance and distances to co-channel interferers change across a cell. The following conservative illustration uses N = 9, n = 2.5, I_0 = 2 and Q = $\sqrt{27}$ = 5.196. If the mobile is at distance r = xR from its serving base station, a nearest-interferer approximation is:

$$SIR(x) \approx (Q - x)^n / (I_0 x^n).$$

Mobile position	**x = r/R**	**Approx. SIR**	**Interpretation**

Near base station	0.1	about 20.0 dB	Strong desired signal; far from cell edge.
Mid-cell	0.5	about 13.8 dB	Typical geometry; serving signal remains dominant.
Cell boundary toward co-channel cell	1.0	about 12.6 dB	Worst geometry; desired path is longest and nearest interferer is closest.

These are geometric estimates, not guaranteed measurements. Shadowing can reduce the desired signal or strengthen an interferer. A log-normal shadowing term $X\sigma$ (dB) is added to received power; a 6–8 dB standard deviation is a reasonable sensitivity scenario for planning, but the actual value must be measured. The lower path-loss exponent $n = 2.5$ produces less attenuation with distance and therefore more co-channel interference. At $n = 4$, distant interferers are attenuated more strongly and SIR improves.

Traffic-time behaviour and blocking

Blocking is governed by offered traffic per sector, not by the mobile's geometric position alone. At 1,000 subscribers, 25 Erlangs over 10 cells and 30 sectors gives only 0.083 Erlang/sector; blocking is negligible. At 20,000 subscribers, 500 Erlangs over 26 cells gives 6.41 Erlangs/sector, approximately the 2% design point. At 100,000 subscribers, 2,500 Erlangs over 126 cells gives 6.61 Erlangs/sector, also approximately 2%.

- Best case: off-peak traffic, mobile near the serving base station, clear line of sight and balanced sector loading. SIR is high and channels are usually available.
- Average case: busy-hour traffic with mobiles spread across sectors. Blocking approaches but should remain at or below 2% after capacity sizing.
- Worst case: cell-edge mobile near a co-channel neighbour during an event surge, with shadowing and a heavily loaded sector. SIR and blocking can both become unacceptable.

A time-of-day traffic map is therefore as important as a coverage map. The design should be checked at busy hour and at known events, not only using daily average traffic.

TASK 5 – Scaling and Sensitivity Analysis (3 Marks)

The principal proportionalities are:

$$\text{Channels per cell} \approx S/N; \qquad Q = \sqrt{(3N)}; \qquad \text{SIR} \propto N^{(n/2)}/I_0.$$

$A_cell \propto R^2$; $M_cover \propto 1/R^2$; $A_total \propto$ subscribers.

$C_system \propto M \times s \times A_B(m_s, 0.02)$.

For fixed N and fixed sector configuration, increasing subscriber count by a factor of 10 requires approximately ten times the traffic capacity. If capacity per cell is held constant, the number of cells must also increase by approximately ten times. In this township the planned cell count rises from a coverage floor of 10 at 1,000 subscribers to 13 at 10,000, 26 at 20,000 and 126 at 100,000.

Subscriber level	**Traffic (Erlangs)**	**Cells planned**	**Network interpretation**
1,000	25	10	Coverage-led macro deployment.
10,000	250	13	Small capacity expansion.
20,000	500	26	Densification needed to protect 2% blocking.
100,000	2,500	126	Dense sectorised macro/micro deployment; site/backhaul cost becomes important.

Path-loss sensitivity: for N = 9 and two dominant sector interferers, the standard SIR estimate is about 14.88 dB at n = 2.5, 18.36 dB at n = 3 and 24.04 dB at n = 4. The conservative edge estimate is about 12.56 dB at n = 2.5 and rises substantially at n = 4. Thus open-campus conditions govern the reuse design. Dense buildings may improve distance attenuation but introduce penetration loss, shadowing and coverage holes.

Break-even judgement: fixed omnidirectional reuse is not practical as the only solution once busy-hour traffic becomes spatially uneven or reaches the 20,000–100,000 range. Sectorisation is preferable for sustained high demand; dynamic borrowing and temporary microcells are preferable for short-lived hotspots. The precise financial break-even point depends on local site and equipment quotations.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

A split replaces one overloaded hexagonal macrocell of radius R by four smaller cells of radius R/2. The total area is approximately preserved because four cells each have one-quarter of the original area. The split creates three net additional cells (one old cell is replaced by four new cells).

Let M_k be the number of cells after k uniform splitting rounds, with M_0 initial cells. Then:

$M_k = 4M_{k-1}$, M_0 = initial number of cells.

If each small cell retains the same channel plan and sector capacity C_cell, the offered capacity is:

$C_k = 4C_{k-1}, \qquad C_0 = M_0 \times C_cell.$

The radius and approximate transmit-power scaling are:

$R_k = R_{k-1}/2 = R_0/2^k; \qquad P_k = P_{k-1}/2^n = P_0/2^{nk}.$

The power relation follows the assignment's stated model $P' = P/2^n$; real radio power planning also depends on antenna gain, receiver sensitivity, indoor penetration and interference targets. Handoff load increases as cells become smaller because mobiles cross cell boundaries more frequently.

Selective splitting

If only a fraction f_k of cells is overloaded in round k, then:

$M_k = M_{k-1}(1 + 3f_k).$

If the overloaded fraction is split, capacity increases by the capacity of three additional cells per split cell:

$C_k = C_{k-1} + 3f_k\, M_{k-1}\, C_cell.$

The work/cost of round k is approximately proportional to $3f_kM_{k-1}$ new sites plus planning, backhaul and neighbour-list updates. Base case: k = 0 is the original macro layout, before any split. The selective recurrence is more economical when demand is concentrated in hostels, libraries or the stadium.

TASK 7 – Solve the Recurrence and Interpret the Result (4 Marks)

For uniform splitting, $M_k = 4M_{k-1}$. Repeated substitution gives $M_1 = 4M_0$, $M_2 = 4(4M_0) = 4^2M_0$, and after k rounds:

$M_k = M_0 \times 4^k; \qquad C_k = C_0 \times 4^k; \qquad R_k = R_0/2^k.$

To carry demand D when initial capacity is C_0, require $C_04^k \geq D$. Dividing by C_0 and taking logarithms:

$4^k \geq D/C_0 \;\Rightarrow\; k \geq \log_4(D/C_0) \;\Rightarrow\; k_min = \text{ceil}[\log_4(D/C_0)].$

For selective splitting with a constant overloaded fraction f, $M_k = M_0(1+3f)^k$. Under the simplified assumption that capacity grows in the same proportion:

$k_min,selective = \text{ceil}[\, \ln(D/C_0) / \ln(1+3f)\,].$

Example: suppose an initial cell has capacity $C_0 = 19.844$ Erlangs and a local hotspot requires D = 250 Erlangs. Uniform splitting requires $\text{ceil}[\log_4(250/19.844)] = 2$ rounds. Capacity becomes $19.844 \times 16 = 317.5$ Erlangs; radius becomes $R_0/4$ and the site count in the affected area becomes 16 times the

original. If only 25% of cells are overloaded in each round, the selective growth factor is 1.75 per round and more rounds are needed, but fewer sites are installed overall.

Power decreases by 2^n per round: at n = 2.5, one round gives $P_1 = P_0/5.657$ and two rounds give $P_2 = P_0/32$. At n = 4, the corresponding factors are 1/16 and 1/256. These are model scalings, not equipment power settings.

Mobility trade-off: halving R approximately doubles the number of cell-boundary crossings per unit travel distance. More cells increase handoff signalling, neighbour relations and risk of ping-pong handoffs. Hysteresis, time-to-trigger and mobility-aware handoff thresholds are needed. Thus the mathematical capacity gain must be balanced against site cost and mobility overhead.

TASK 8 – Implement and Test the Planning Strategies (4 Marks)

The following Python reference implementation reproduces the main analytical planning calculations and provides a Monte-Carlo SIR experiment. It uses a fixed random seed for reproducibility. The strategy modes are: (1) fixed reuse with six dominant interferers, (2) sectorised reuse with two dominant interferers, and (3) dynamic borrowing represented by an optional traffic-capacity relief factor while retaining the same SIR guard. For a deployment-grade simulator, replace the simplified interferer geometry with measured site coordinates, antenna patterns and a full hex-grid neighbour map.

```
import math
import numpy as np

TOTAL_CHANNELS = 336
BLOCKING_TARGET = 0.02
TRAFFIC_PER_USER = 0.025
SEED = 42

def erlang_b(channels, offered):
    b = 1.0
    for m in range(1, channels + 1):
        b = offered * b / (m + offered * b)
    return b

def erlang_capacity(channels, target=BLOCKING_TARGET):
    lo, hi = 0.0, channels * 3.0
```

```
    for _ in range(100):
        mid = (lo + hi) / 2
        if erlang_b(channels, mid) > target:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2

def cluster_size(i, j):
    return i*i + i*j + j*j

def sir_db(N, n, interferers=6, edge=False):
    q = math.sqrt(3*N)
    ratio = (q - 1) if edge else q
    return 10 * math.log10((ratio**n) / interferers)

def plan(N=9, sectors=3, subscribers=100000):
    channels_per_sector = (TOTAL_CHANNELS // N) // sectors
    cap_sector = erlang_capacity(channels_per_sector)
    cap_cell = sectors * cap_sector
    demand = subscribers * TRAFFIC_PER_USER
    cells = math.ceil(demand / cap_cell)
    radius = math.sqrt(12 / (2.598 * cells))
    return demand, channels_per_sector, cap_sector, cap_cell, cells, radius

def monte_carlo_sir(N=9, n=2.5, sectors=True, samples=100000):
    rng = np.random.default_rng(SEED)
    I0 = 2 if sectors else 6
    q = math.sqrt(3*N)
    # Uniform user locations in a circular approximation to a cell.
    r = np.sqrt(rng.random(samples))
    # First-tier interferer distances are represented by a conservative
    # nearest-interferer distance, with random angular/position variation.
```

```
    angle = rng.uniform(0, 2*np.pi, samples)
    d = np.sqrt(q*q + r*r - 2*q*r*np.cos(angle))
    # Small log-normal shadowing perturbation (6 dB standard deviation).
    shadow = rng.normal(0, 6, samples)
    sir = 10*np.log10((d**(-n) / (I0 * (r + 0.03)**(-n)))) + shadow
    return np.mean(sir), np.percentile(sir, 5), np.mean(sir < 12)

for users in [1000, 10000, 20000, 100000]:
    print(users, plan(N=9, sectors=3, subscribers=users))

for sectors in [False, True]:
    print("sectorised =", sectors,
          monte_carlo_sir(N=9, n=2.5, sectors=sectors))
```

Test matrix required by the assignment: N = 3, 4, 7, 12; n = 2.5, 3, 4; omnidirectional and three-sector layouts; subscriber loads 1,000, 10,000, 20,000 and 100,000. For each run, store the seed, cluster size, path-loss exponent, sector setting, channel count, mean SIR, 5th-percentile SIR, outage probability, Erlang-B blocking, cells and execution time.

Sanity check: for N = 7, n = 4 and six equidistant interferers, the analytical expression gives 18.66 dB, matching the assignment's stated value of approximately 18.7 dB. The code is an executable reference; Monte-Carlo output must be generated in the user's Python environment before it is described as an observed experimental result.

TASK 9 – Record and Present Experimental Performance (3 Marks)

A fair comparison must use the same subscriber loads, random seed, path-loss exponent and propagation assumptions for every strategy. Record the following measurements:

- SIR distribution: mean SIR, 5th-percentile SIR and outage probability P(SIR < 12 dB).
- Capacity: channels per cell/sector, Erlang-B blocking probability, supported Erlangs and maximum subscribers at 0.025 Erlang each.
- Deployment: number of sites, cell radius, reuse distance, sectors and temporary sites.
- Efficiency: planning cost, operating cost, handoff rate and simulation runtime.

Measurement	Fixed omni reuse	Sectorised reuse	Dynamic / hierarchical
SIR model	$I_0 \approx 6$	$I_0 \approx 2$	$I_0 \approx 2$, with channel guard check
N used for comparison	9	9	9 base layer; micro layer at hotspots
Channels per sector	37 per cell, omni	12	12 home channels + safe borrowed channels
2% blocking capacity/cell	28.25 E/cell (37-channel pool)	19.84 E/cell (3 × 12 channels)	At least 19.84 E/cell; borrowing can reduce transient blocking if safe
Main weakness	Fails open-campus SIR target in simple model	More channels split among sectors; antenna complexity	Coordination complexity and interference risk
Sites at 100,000 users	Depends on omni SIR-feasible N; N=9 fails at n=2.5	126	Approximately 126 base cells plus hotspot overlays

Recommended plots: (1) SIR cumulative distribution function (CDF) for each strategy; (2) blocking probability versus offered load; (3) required cells versus subscriber count; (4) capacity versus cluster size N; and (5) cost per supported Erlang versus number of sites. Do not invent plotted measurements: the tables above are analytical sizing values, while Monte-Carlo curves should be exported from the supplied code after execution.

TASK 10 – Compare Theory with Simulation (4 Marks)

The analytical model predicts mean/idealised SIR and Erlang-B blocking under simplified assumptions. A simulation should agree in trend and approximate scale, but exact values will differ because it includes user position, random shadowing and finite network boundaries.

Comparison item	Analytical prediction	Why simulation may differ
SIR	$SIR \approx Q^n/I_0$; conservative edge model uses $(Q-1)^n/I_0$.	Interferers are not exactly equidistant; antenna sidelobes, terrain, shadowing and user position change received powers.

Blocking	Erlang-B with fixed channel count and offered traffic.	Real traffic may be bursty, arrivals may not be Poisson, calls may have different holding times and sectors may be imbalanced.
Number of cells	Ceiling of traffic/capacity and area/hexagon.	Real sites cannot be placed on a perfect grid; buildings, roads, permissions and backhaul constrain locations.
Reuse and boundaries	Infinite repeating hexagonal pattern.	A finite 12 km² township has fewer interferers near its boundary and may need external neighbour coordination.
Channel allocation	Integer channels per cluster/sector.	Rounding, guard channels, control resources and dynamic borrowing change the usable pool.

The N = 7, n = 4, I_0 = 6 check gives 18.66 dB, which verifies the formula against the assignment's reference value. For the selected N = 9 sectorised design, the standard SIR approximation is 14.88 dB and the conservative cell-edge estimate is 12.56 dB at n = 2.5. The latter is the design acceptance value. The Erlang-B capacity of 12 channels per sector is 6.615 Erlangs at 2% blocking. If a simulation produces a 5th-percentile SIR below 12 dB or blocking above 2%, the design must be revised rather than reporting the analytical estimate as a measured success.

Based on the analytical results, sectorised reuse scales better than fixed omnidirectional N = 9 in the open-campus case because it reduces the dominant interferer count. Dynamic borrowing is useful for temporary traffic imbalance but cannot create unlimited capacity; it must be backed by additional microcells when all neighbouring sectors are busy.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommended township design

Deploy a three-sector, 120° sectorised cellular network with N = 9. Use 12 traffic channels per sector, retaining the remaining channel in the nominal 37-channel cell allocation as a spare/guard resource. Set the minimum design SIR to 12 dB. Plan the radio network against n = 2.5 because this is the most interference-challenging value in the supplied range; use n = 3 for typical planning and n = 4 for dense built-up sensitivity checks.

Subscriber forecast	Offered traffic	Cells	Approx. cell radius	Capacity check
1,000	25 E	10	0.679 km	Coverage-led; low load
10,000	250 E	13	0.596 km	13 × 19.844 = 258 E
20,000	500 E	26	0.421 km	26 × 19.844 = 516 E
100,000	2,500 E	126	0.191 km	126 × 19.844 = 2,500.3 E

The 100,000-subscriber result is close to the 2% blocking design limit, so operational planning should include a small capacity reserve where site permissions and spectrum reuse allow. The calculated edge SIR at n = 2.5 is approximately 12.56 dB, above the 12 dB threshold by only 0.56 dB. A drive test is essential; where shadowing or antenna imperfections erase this margin, use local microcells, antenna optimisation or a more conservative local reuse plan.

Why this is preferred

- Compared with fixed omnidirectional N = 9, three-sector reuse reduces dominant first-tier interference and meets the open-campus SIR target in the simplified edge model.
- Compared with increasing N substantially, N = 9 retains more channels per sector and avoids excessive site density.
- Compared with dynamic borrowing alone, sectorisation provides permanent interference control; borrowing is added as a supplementary load-balancing feature.
- The staged site count follows demand and avoids installing 126 sites at the beginning when only 1,000 subscribers are present.

Cost and limitations: the main cost drivers are site acquisition, radio/antenna equipment, power, backhaul, survey and maintenance. Sectorisation increases antenna and optimisation complexity but avoids purchasing more spectrum. Cell splitting increases capacity and reduces transmit power, but increases handoffs and operating cost. The design remains technically scalable beyond 100,000 subscribers only through further densification, microcells and traffic engineering; the 336-channel allocation remains a hard resource constraint.

TASK 12 – Re-evaluate and Adapt the Design – Day-4 Challenge (8 Marks)

A. Impact analysis (2 marks)

The event brings 25,000 attendees, each generating 0.05 Erlang. Therefore:

$A_{event} = 25{,}000 \times 0.05 = 1{,}250$ Erlangs in 1 km² for four hours.

The event traffic density is 1,250 Erlangs/km². For comparison, the 100,000-subscriber township-wide demand is 2,500 Erlangs/12 km² = 208.3 Erlangs/km² on average. The event-zone density is therefore about six times the township average. It also doubles the traffic per attendee compared with the normal 0.025-Erlang planning assumption.

The assumptions of spatially uniform traffic, steady busy-hour demand and permanent capacity no longer hold. However, the surge is predictable, localised and lasts only four hours, occurring roughly twice per month. This strongly favours a temporary overlay rather than permanently installing enough capacity for an occasional peak.

B. Evaluation of the original design (2 marks)

At 100,000 subscribers, 126 cells cover 12 km², or about 10.5 cells per km². A 1 km² stadium zone therefore contains roughly 10–11 cells on average. If 1,250 Erlangs are concentrated in 10.5 cells, the average is approximately 119 Erlangs per cell, or 39.7 Erlangs per sector. Each 12-channel sector is designed for only 6.615 Erlangs at 2% blocking. Erlang-B blocking at approximately 39.7 Erlangs with 12 channels is extremely high (well above 70%).

The original design cannot be reused unchanged. Blocking occurs first because the stadium sectors exhaust their traffic channels. Interference may also rise if uncontrolled borrowing is attempted from nearby cells. The design's normal SIR result does not guarantee event capacity.

C. Adaptation or redesign (2 marks)

Decision: MODIFY TEMPORARILY. Keep the permanent N = 9, three-sector macro network for normal days. For the event, deploy approximately 63 temporary low-power microcells / cells-on-wheels across the 1 km² stadium zone. At 19.844 Erlangs per three-sector cell, the minimum capacity count is:

$M_{event} = \text{ceil}(1{,}250 / 19.844) = 63$ temporary cells.

The corresponding idealised microcell radius is:

$R_{event} = \sqrt{[1 / (63 \times 2.598)]} \approx 0.078$ km (78 metres).

Each temporary cell uses the same three-sector concept and N = 9 reuse discipline, with transmit power reduced for the smaller radius. The reuse distance is D = 5.196 × 0.078 ≈ 0.406 km. A detailed site plan must check actual stadium dimensions, crowd distribution, antenna direction and co-channel neighbours.

Use dynamic channel borrowing only when the donor sector remains below its blocking target and the SIR check passes. Reserve handoff/guard resources for users moving between temporary and permanent cells.

The temporary overlay increases local capacity by about 63 × 19.844 = 1,250 Erlangs, enough for the stated event load at the 2% design point. It avoids permanent construction of 63 extra sites. Costs include temporary radio rental, transport, setup, backhaul, power, event-day monitoring and post-event removal.

D. Evidence-based defence (2 marks)

A permanent redesign would pay recurring site, power, rent and maintenance costs for a peak that occurs only about twice a month and lasts four hours. Retaining the original network unchanged would cause severe blocking in the stadium. A temporary microcell overlay is the balanced choice: it adds capacity where and when needed, maintains the fixed spectrum through spatial reuse, and can be removed after the event.

The decision is supported by the traffic calculation (1,250 Erlangs), the comparison with the 2% per-sector capacity (6.615 Erlangs), the estimated 63-cell requirement, and the short predictable event duration. Final deployment must be validated by a venue walk-test and a site-specific Monte-Carlo simulation. If the event footprint or crowd distribution is uneven, add temporary cells to the most heavily loaded stands and entrances first.

Overall Conclusion

The assignment demonstrates that cellular capacity is not determined by the number of licensed channels alone. It depends on the reuse cluster, cell radius, antenna sectorisation, traffic distribution, interference and blocking target. The selected N = 9, three-sector design uses the fixed 336-channel allocation efficiently and provides a scalable path from 1,000 to 100,000 subscribers through staged densification. For the short stadium surge, temporary microcells are more economical than permanent township-wide redesign.